DISCRETE MATHEMATICS AND GRAPH THEORY

Complete Topic-Wise Notes

Simple English | Definitions | Formulas | Examples | Exam Revision

For BCA / MCA / Computer Science Students

This guide covers common topics. Exact units and notation may differ by university, so compare it with your official syllabus.

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1. Mathematical Logic and Propositions

Proposition: a declarative statement that is either true or false, but not both. Example: “7 is prime” is true. A question or command is not a proposition.

Logical connectives

- **Negation (¬p):** not p; reverses the truth value.
- **Conjunction (p ∧ q):** p AND q; true only when both are true.
- **Disjunction (p ∨ q):** inclusive OR; true when at least one is true.
- **Exclusive OR (p ⊕ q):** true when exactly one proposition is true.
- **Implication (p → q):** false only when p is true and q is false.
- **Biconditional (p ↔ q):** true when p and q have the same truth value.

Truth table

p	q	p∧q	p∨q	p→q	p↔q
T	T	T	T	T	T
T	F	F	T	F	F
F	T	F	T	T	F
F	F	F	F	T	T

Tautology, contradiction and contingency

- Tautology: always true, e.g., p ∨ ¬p.
- Contradiction: always false, e.g., p ∧ ¬p.
- Contingency: true for some truth assignments and false for others.

2. Logical Equivalences and Inference Rules

Important logical laws

- Double negation: $\neg(\neg p) \equiv p$.
- De Morgan's laws: $\neg(p \wedge q) \equiv \neg p \vee \neg q$; $\neg(p \vee q) \equiv \neg p \wedge \neg q$.
- Implication: $p \to q \equiv \neg p \vee q$.
- Contrapositive: $p \to q \equiv \neg q \to \neg p$.
- Commutative: $p \wedge q \equiv q \wedge p$; $p \vee q \equiv q \vee p$.
- Associative: $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$, and similarly for $\vee$.
- Distributive: $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$; dual law also holds.
- Identity: $p \wedge T \equiv p$; $p \vee F \equiv p$.
- Domination: $p \vee T \equiv T$; $p \wedge F \equiv F$.
- Idempotent: $p \wedge p \equiv p$; $p \vee p \equiv p$.
- Absorption: $p \vee (p \wedge q) \equiv p$; $p \wedge (p \vee q) \equiv p$.

Rules of inference

- Modus Ponens: p, $p \to q$ therefore q.
- Modus Tollens: $\neg q$, $p \to q$ therefore $\neg p$.
- Hypothetical Syllogism: $p \to q$, $q \to r$ therefore $p \to r$.
- Disjunctive Syllogism: $p \vee q$, $\neg p$ therefore q.
- Addition: p therefore $p \vee q$.
- Simplification: $p \wedge q$ therefore p.
- Conjunction: p and q therefore $p \wedge q$.

3. Predicates and Quantifiers

A predicate is a statement involving variable(s) whose truth depends on the values assigned to them. Example: P(x): x > 5.

- Universal quantifier ∀x P(x): P(x) is true for every x in the domain.
- Existential quantifier ∃x P(x): P(x) is true for at least one x in the domain.
- Negation: ¬∀x P(x) ≡ ∃x ¬P(x).
- Negation: ¬∃x P(x) ≡ ∀x ¬P(x).
- Order matters: ∀x∃y P(x,y) need not mean the same as ∃y∀x P(x,y).
- Free variable: not bound by a quantifier; bound variable: governed by a quantifier.

Example

“Every student passed” can be written ∀x(Student(x) → Passed(x)). “Some student passed” can be written ∃x(Student(x) ∧ Passed(x)).

4. Set Theory and Set Operations

A set is a well-defined collection of distinct objects. Example: A = {1,2,3}.

Basic terms

- Element: $x \in A$ means x belongs to A; $x \notin A$ means it does not.
- Subset: $A \subseteq B$ if every element of A is in B.
- Proper subset: $A \subset B$ when $A \subseteq B$ and $A \neq B$.
- Empty set: $\varnothing$ has no elements; universal set U contains the objects under discussion.
- Power set P(A): set of all subsets of A. If |A|=n, then |P(A)|=2■.
- Cardinality |A|: number of elements in a finite set.

Set operations

- Union $A \cup B$: elements in A or B or both.
- Intersection $A \cap B$: elements common to A and B.
- Difference A–B: elements in A but not B.
- Complement A■: elements in U but not A.
- Symmetric difference A■B: elements in exactly one of A and B.
- Cartesian product $A \times B$: ordered pairs (a,b) with $a \in A$ and $b \in B$.

Set laws and counting

- Commutative: $A \cup B = B \cup A$; $A \cap B = B \cap A$.
- Associative and distributive laws apply to $\cup$ and $\cap$.
- De Morgan: (A∪B)■=A■∩B■; (A∩B)■=A■∪B■.
- Inclusion-exclusion: $|A \cup B| = |A| + |B| - |A \cap B|$.
- For three sets: $|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$.

5. Relations and Their Properties

A binary relation R from A to B is a subset of A×B. A relation on A is a subset of A×A.

Properties of a relation on A

- Reflexive: $(a,a) \in R$ for every $a \in A$.
- Irreflexive: $(a,a) \notin R$ for every $a \in A$.
- Symmetric: if aRb, then bRa.
- Antisymmetric: if aRb and bRa, then a=b.
- Asymmetric: if aRb, then bRa is false; implies irreflexivity.
- Transitive: if aRb and bRc, then aRc.

Representations

- Ordered pairs list the relation directly.
- A matrix uses 1 for a related pair and 0 otherwise, with a chosen ordering of the set.
- A directed graph draws an arrow $a \to b$ when aRb.

Inverse and composition

- Inverse relation R^{-1} reverses every ordered pair: $(a,b) \in R^{-1}$ iff $(b,a) \in R$.
- Composition $S \circ R$ relates a to c when there exists b such that aRb and bSc; notation order may vary by textbook.

6. Equivalence Relations and Partial Orders

Equivalence relation

A relation is an equivalence relation if it is reflexive, symmetric and transitive.

- Equivalence class [a] contains all elements related to a.
- Equivalence classes partition the set into disjoint non-empty groups.
- Example: congruence modulo n on integers: $a \equiv b \pmod{n}$ when n divides a–b.

Partial order

A relation is a partial order if it is reflexive, antisymmetric and transitive. A set with a partial order is a poset.

- Comparable elements: $a \leq b$ or $b \leq a$.
- Total/linear order: every pair of elements is comparable.
- Hasse diagram: simplified diagram of a finite poset; omit loops, transitive edges and arrowheads (higher elements are usually drawn above).
- Minimal element: no strictly smaller element; least element is below every element. These are not always the same.
- Maximal element: no strictly larger element; greatest element is above every element.

Lattices

A lattice is a poset in which every pair of elements has a greatest lower bound (meet) and a least upper bound (join).

7. Functions

A function f:A→B assigns exactly one output in B to every input in A. A is the domain and B is the codomain.

- Image/range: outputs actually attained by f; range is a subset of codomain.
- Injective (one-to-one): f(a)=f(b) implies a=b.
- Surjective (onto): every element of codomain is hit.
- Bijective: both injective and surjective; has an inverse function.
- Composition: (g■f)(x)=g(f(x)).
- Identity function: idA(x)=x.
- Inverse function f■$^{-1}$ exists as a function from B to A when f is bijective.
- Pigeonhole consequence: no injection from a larger finite set to a smaller finite set.

8. Mathematical Induction and Recurrence Relations

Mathematical induction

- Base case: prove the statement for the starting integer.
- Inductive hypothesis: assume it holds for n=k.
- Inductive step: use the hypothesis to prove it for n=k+1.
- Conclusion: the statement holds for all integers in the specified range.

Strong induction

Assume the statement is true for all values from the starting point through k, then prove it for k+1.

Recurrence relations

A recurrence defines terms of a sequence using earlier terms, plus initial conditions.

- Example: a■=a■■■+2 with a■=1 gives 1,3,5,7,…
- Arithmetic sequence: a■=a■+(n–1)d.
- Geometric sequence: a■=a■r■■¹.
- For linear homogeneous recurrences with constant coefficients, try a characteristic equation; repeated roots require additional polynomial factors.

9. Counting Principles and Pigeonhole Principle

- Addition rule: choose one of disjoint alternatives with m or n choices → m+n choices.
- Multiplication rule: a sequence of choices with m then n possibilities → mn possibilities.
- Complement rule: count desired cases as total cases minus undesired cases.
- Pigeonhole principle: placing n+1 objects into n boxes puts at least two in one box.
- Generalized form: placing N objects into k boxes means some box contains at least ■N/k■ objects.
- Inclusion-exclusion avoids double-counting overlapping sets.

10. Permutations and Combinations

- Factorial: n!=n(n–1)…2·1; 0!=1.
- Permutation of r from n distinct objects: P(n,r)=n!/(n–r)!.
- Combination: C(n,r)=n!/[r!(n–r)!]. Order matters for permutations; not for combinations.
- Permutations of n objects with repeated groups of sizes n■,…,n■: n!/(n■!…n■!).
- Circular arrangements of n distinct objects, when rotations are equivalent: (n–1)!.
- Binomial theorem: (x+y)■=Σ■■■■ C(n,r)x■■■y■.
- C(n,r)=C(n,n–r); C(n,r)=C(n–1,r)+C(n–1,r–1).

11. Probability Basics (if included)

- For equally likely outcomes: $P(E)=|E|/|S|$.
- $0 \le P(E) \le 1$; $P(S)=1$; $P(\varnothing)=0$.
- Complement: $P(E')=1-P(E)$.
- Addition rule: $P(A \cup B)=P(A)+P(B)-P(A \cap B)$.
- Conditional probability: $P(A|B)=P(A \cap B)/P(B)$, when $P(B)>0$.
- Independence: $P(A \cap B)=P(A)P(B)$.

12. Graph Theory: Basics and Terminology

A graph G=(V,E) consists of a vertex set V and an edge set E connecting vertices.

- Vertex/node: object in the graph.
- Edge: connection between vertices.
- Adjacent vertices share an edge; incident edge touches a vertex.
- Order of graph: |V|; size: |E|.
- Loop: edge from a vertex to itself.
- Parallel/multiple edges: more than one edge joining the same pair (allowed in multigraphs).
- Simple graph: no loops or parallel edges.
- Isolated vertex: degree 0; pendant/leaf vertex: degree 1.
- Subgraph: graph formed from subsets of vertices and edges.

Common graph types

- Undirected graph: edges have no direction.
- Directed graph (digraph): edges/arcs have direction.
- Weighted graph: edges carry weights/costs.
- Complete graph K■: every pair of distinct vertices is adjacent.
- Null graph: vertices but no edges.
- Regular graph: all vertices have equal degree.
- Bipartite graph: vertices can be divided into two sets with edges only between the sets.
- Complete bipartite graph K■,■ connects every vertex in one part to every vertex in the other.

13. Degree and Handshaking Lemma

- Degree deg(v): number of edge-ends incident to v; a loop counts twice in an undirected graph.
- Handshaking lemma: Σ■ deg(v)=2|E| for any finite undirected graph.
- Therefore, the number of odd-degree vertices is always even.
- In a directed graph, indegree counts incoming arcs and outdegree counts outgoing arcs.
- Σ indegree(v)=Σ outdegree(v)=|E| for a finite digraph.
- Complete graph: |E(K■)|=n(n−1)/2; every vertex has degree n−1.
- Complete bipartite graph: |E(K■,■)|=mn.
- Cycle C■ has n edges and each vertex has degree 2; path P■ has n vertices and n−1 edges.

14. Walks, Trails, Paths and Connectivity

- Walk: sequence of vertices/edges where consecutive vertices are connected; repetitions may occur.
- Trail: walk with no repeated edge.
- Path: commonly a walk with no repeated vertex (some textbooks use variants; follow your course definition).
- Closed walk starts and ends at the same vertex.
- Circuit: closed trail.
- Cycle: closed path with no repeated vertices except the start/end.
- Connected graph: a path exists between every pair of vertices.
- Connected component: maximal connected subgraph.
- Bridge/cut-edge: removing it increases the number of connected components.
- Cut-vertex/articulation point: removing it increases the number of components.

15. Euler and Hamilton Graphs

Euler

- Euler trail: traverses every edge exactly once.
- Euler circuit: Euler trail that starts and ends at the same vertex.
- For a connected undirected graph (ignoring isolated vertices), an Euler circuit exists iff every vertex has even degree.
- An Euler trail but not circuit exists iff exactly two vertices have odd degree.
- For directed graphs, degree-balance and connectivity conditions apply; use the exact theorem taught in your course.

Hamilton

- Hamilton path visits every vertex exactly once.
- Hamilton cycle returns to the starting vertex after visiting every vertex exactly once.
- There is no simple degree-only test that characterizes all Hamiltonian graphs.
- Necessary conditions and sufficient theorems such as Dirac's theorem can help, but must be applied with their assumptions.

Key difference

Euler problems focus on using every edge once; Hamilton problems focus on visiting every vertex once.

16. Trees and Spanning Trees

A tree is a connected undirected graph with no cycles.

- A tree with n vertices has exactly n−1 edges.
- In a tree, there is exactly one simple path between every pair of vertices.
- Removing any edge disconnects a tree; adding an edge creates a cycle.
- A forest is an acyclic graph; each component is a tree.
- A spanning tree of a connected graph includes all vertices and is a tree subgraph.
- A graph can have more than one spanning tree.
- A rooted tree identifies a root; parent, child, sibling, ancestor and descendant describe relationships.
- Binary tree: each node has at most two children.

Minimum spanning tree (MST)

An MST is a spanning tree of a connected weighted undirected graph with minimum total edge weight.

- Kruskal's algorithm: sort edges by increasing weight; add an edge if it does not create a cycle.
- Prim's algorithm: grow one tree by repeatedly adding the lowest-weight edge connecting the tree to a new vertex.
- If edge weights are distinct, the MST is unique; with ties, multiple MSTs may exist.

17. Planar Graphs

A planar graph can be drawn in the plane without edge crossings, except at shared endpoints.

- A planar embedding divides the plane into faces, including the unbounded outer face.
- Euler's formula for a connected planar graph: $V - E + F = 2$.
- For a connected planar simple graph with $V \geq 3$: $E \leq 3V - 6$.
- For a connected planar simple bipartite graph with $V \geq 3$: $E \leq 2V - 4$.
- $K_{\blacksquare}$ and $K_{\blacksquare,\blacksquare}$ are non-planar.
- These edge bounds are useful for proving non-planarity; they are not sufficient on their own to prove planarity.

18. Graph Coloring

- A proper vertex coloring assigns colors so adjacent vertices have different colors.
- Chromatic number $\chi(G)$: minimum number of colors needed for a proper vertex coloring.
- A nonempty edgeless graph has chromatic number 1; a complete graph K_n has $\chi(K_n)=n$.
- A graph is bipartite iff it has no odd cycle; a graph with at least one edge that is bipartite needs 2 colors.
- A cycle C_n needs 2 colors if n is even and 3 if n is odd.
- Planar graphs can be colored with at most 4 colors (Four Color Theorem).
- Edge coloring requires incident edges to receive different colors; its minimum is the chromatic index.

19. Shortest Paths and Minimum Spanning Trees

Shortest path

- Shortest path minimizes total edge weight between two vertices.
- BFS finds shortest paths by number of edges in an unweighted graph.
- Dijkstra's algorithm works with nonnegative edge weights.
- Bellman–Ford supports negative edge weights and can detect reachable negative-weight cycles.
- Floyd–Warshall computes all-pairs shortest paths using dynamic programming.

MST vs shortest path tree

A minimum spanning tree minimizes the total weight connecting all vertices. A shortest path tree minimizes distances from a chosen source. They are different optimization goals.

20. Directed Graphs and Graph Representations

- Adjacency matrix: n×n matrix; entry indicates whether an edge from i to j exists (or stores its weight).
- Adjacency list: each vertex stores its neighbors; often space-efficient for sparse graphs.
- Incidence matrix: rows represent vertices and columns represent edges.
- In a directed graph, a source has indegree 0 and a sink has outdegree 0.
- A directed graph is strongly connected if each vertex can reach every other vertex.
- A DAG is a directed acyclic graph.
- Topological ordering of a DAG places u before v for every edge u→v; it can be found by Kahn's algorithm or DFS.

Graph traversal

- BFS (Breadth-First Search): explores layer by layer using a queue.
- DFS (Depth-First Search): explores deeply before backtracking, using recursion or a stack.
- For adjacency-list representations, BFS and DFS run in O(V+E) time.

21. Boolean Algebra and Lattices (where included)

- Boolean variables take values 0 and 1.
- OR corresponds to join, AND to meet, and NOT to complement in Boolean algebra.
- Identity: $x \vee 0 = x$; $x \wedge 1 = x$.
- Complement: $x \vee \neg x = 1$; $x \wedge \neg x = 0$.
- Idempotent: $x \vee x = x$; $x \wedge x = x$.
- De Morgan: $\neg(x \vee y) = \neg x \wedge \neg y$; $\neg(x \wedge y) = \neg x \vee \neg y$.
- A Boolean algebra is a complemented distributive lattice with least and greatest elements.

22. Algebraic Structures (where included)

- Semigroup: set with an associative binary operation.
- Monoid: semigroup with an identity element.
- Group: monoid in which every element has an inverse; equivalently, associative operation, identity and inverses.
- Abelian group: group whose operation is commutative.
- Ring: set with addition and multiplication satisfying specified axioms; addition forms an abelian group and multiplication is associative, with distributive laws.
- Field: commutative ring with identity where every nonzero element has a multiplicative inverse.

Check your course's exact axioms and conventions; some textbooks define ring identity requirements differently.

23. Formula Sheet and Exam Questions

Quick formulas

- |P(A)|=2■ when |A|=n.
- |A∪B|=|A|+|B|−|A∩B|.
- P(n,r)=n!/(n−r)!; C(n,r)=n!/[r!(n−r)!].
- Σ deg(v)=2E; odd-degree vertex count is even.
- |E(K■)|=n(n−1)/2; |E(K■,■)|=mn.
- Tree with n vertices: E=n−1.
- Connected planar graph: V−E+F=2.
- Simple planar graph V≥3: E≤3V−6.
- Bipartite simple planar graph V≥3: E≤2V−4.

Important exam questions

- Construct truth tables and prove logical equivalences.
- Negate quantified statements and identify valid inference rules.
- Solve set identities and inclusion-exclusion problems.
- Test whether a relation is reflexive, symmetric, antisymmetric and transitive.
- Determine equivalence classes, posets and Hasse diagrams.
- Test whether a function is injective, surjective or bijective.
- Prove a formula by mathematical induction.
- Solve permutation, combination and pigeonhole problems.
- Find degrees, paths, components and graph representations.
- Determine whether a graph has an Euler trail/circuit or Hamilton path/cycle.
- Find spanning trees and MSTs using Kruskal's and Prim's algorithms.
- Use Euler's formula and edge bounds to test planarity.
- Find chromatic number for standard graphs.
- Apply BFS, DFS, Dijkstra, Bellman–Ford or topological sorting where included.

Study method

For each topic, learn the definition, write the key formula, solve at least two examples, and practise past-paper questions. Use your official university syllabus to confirm which optional topics (probability, lattices, Boolean algebra and algebraic structures) are included.